Lecture 13

Image Analysis:
Measuring size and shape of objects through imaging

Contents

What is Size

- Definition (after Webster)
  - The quality of an object that determines how much space it occupies
  - Dimensions or Magnitude
- D'Arcy Thompson Wentworth
  - "On growth and form", 1917
  - "A *form* of an object is defined when we know its magnitude, actual or relative, in various directions."

Magnitude

- Refers to
  - A small elephant
  - A large rat
- For measurement
  - Compare magnitudes
  - Compare at different order of magnitude
- Use different Magnitude/Size features
  - Compare within a class
  - Compare between classes
- Magnitude/Size features: *Shape*

What is Shape

- Definition (after Webster)
  - The quality of an object that depends on the relative position of all points on its surface
- The boundary of the body describes a shape
  - How can we measure features on the boundary of the shape
  - What features describe a shape best
  - Depending on type of object: feature selection

Size and Shape

- Spatial occupancy related to *Size*
  - Length
  - Area
  - Volume
  - Density: Average density, Optical density
  - Moments
- Boundary occupancy, behavior related to *Shape*
  - Shape factor, Compactness
  - Convexity
  - Curvature
  - Bending energy
  - Moments

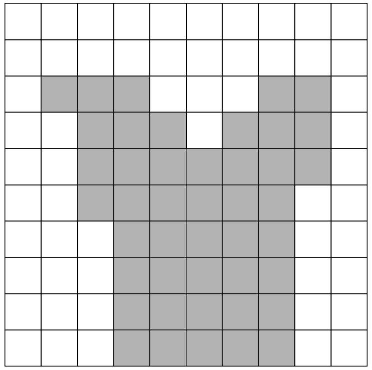
Expressing Size & Shape

- In the 2D/3D case
  - Pixels/ Voxels
- Generalized
  - Imels
- Calibration
  - Imel units to Metric units
- Measurement
  - Set of observations
  - Variance and standard deviation

Surface Area

- Area can be expressed analytically by a number of formulas:
  - Circle $A = \pi \cdot r^2$
  - Triangle $A = \frac{1}{2} a \cdot h$
- Given a shape on a discrete grid, how should the area be computed.
  - # number of pixels
  - express area in imel or SI

Calculate Surface Area



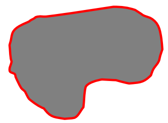
0
0
5
6
7
6
5
5
5
5

Pxy-tables

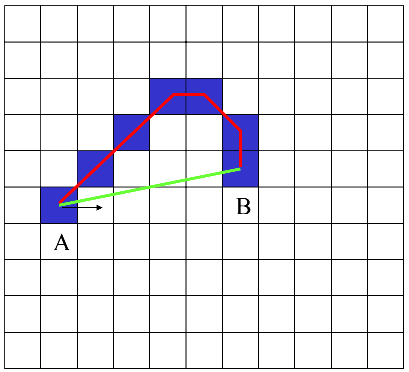
Area = 44

Perimeter

- Perimeter is a measure that is derived from the boundary pixels. But how?
  - Accurate estimator
  - Grid, assume 8-connected contour
- Perimeter of the object is computed from the contour derived from the object.



Simple line on Grid



- Euclidean Distance
  - D(A,B) = 5,1
- Distance along Contour
  - chaincode
  - cc(A,B) = 1,1,1,0,7,6
  - length measurement from cc(A,B)
  - requires Estimator L

Estimating Perimeter

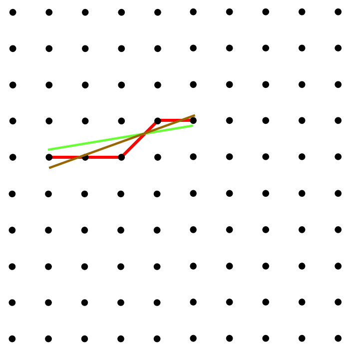
- Perimeter is estimated from ChainCode
- Perimeter is expressed in pixels per unit length
- N_e: # even chaincodes
  - 4 connected, city-block
- N_o: # odd chaincodes
  - 8-connected, chessboard
- Nc: # corners
  - chaincode changes direction
- Generic form: $L_{vs} = \alpha \cdot N_e + \beta \cdot N_o - \gamma \cdot N_c$

3	2	1
4	P(n,m)	0
5	6	7

8-connected

Line on a Grid

- Sampling
  - Effect digitization
- Probability
  - Distribution
  - Variation in digitization



Estimators for Perimeter

- Simple

$$L_0 = N_e + N_o$$

- (1970) Freeman

$$L_f = (1.0) \cdot N_e + (\sqrt{2}) \cdot N_o$$

- (1978) Groen & Verbeek

$$L_{gv} = (1.059) \cdot N_e + (1.183) \cdot N_o$$

- (1979) Profitt & Rosen

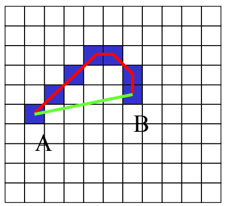
$$L_{pr} = (0.984) \cdot N_e + (1.340) \cdot N_o$$

- (1982) Vossepoel & Smeulders, a.ka. Corner-count

$$L_{vs} = (0.980) \cdot N_e + (1.406) \cdot N_o - (0.091) \cdot N_c$$

Results Estimators

- cc(A,B) = 1,1,1,0,7,6
  - $N_e = 2$, $N_o = 4$, $N_c=3$
- Simple: L = 2 + 4 = 6
- Freeman: L = 2 + 4*sqrt(2) = 7.66
- Groen: L = 2.118 + 4.732 = 6.85
- Profitt: L = 1.968 + 5.36 =7.328
- Vossepoel: L= 1.96 + 5.624 – 0.273 = 7.311

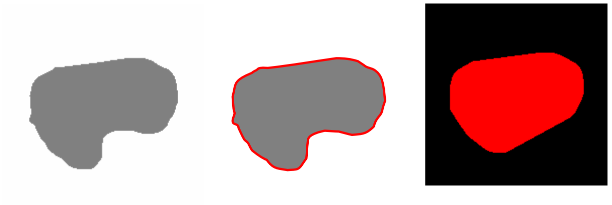


- Unbiased Estimator E(t) = s,
  - E(t) – s = b (b = bias)
  - E(t) – s = 0 unbiased
  - Length: n larger, then b smaller: approx 0 (asymptotic)
  - Bias corner count for n > 100, smaller then 0.08%

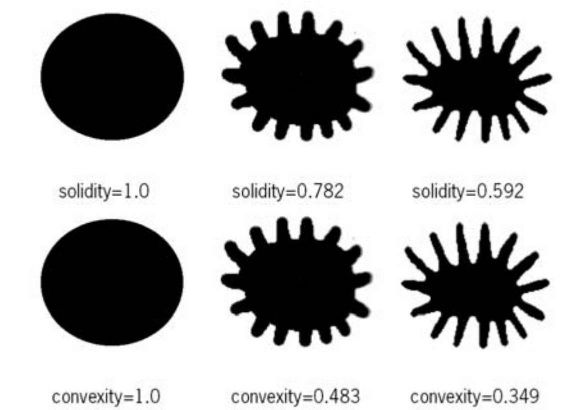
Using Convex Hull

- Ratio between convex hull of an object and the object
  - Measured as ratio of area
    - $0 < A_{object} / Area_{hull} <= 1$
    - $Area_{hull} >= A_{object}$
    - Note: A_{object} is space taken by object (filled)
    - Ratio $A_{object} / Area_{hull}$: *Solidity*
  - Measured as ratio of perimeter
    - $0 < Perimeter_{hull} / Perimeter_{object} <=1$
    - $Perimeter_{object} >= Perimeter_{hull}$
    - Ratio $Perimeter_{hull} / Perimeter_{object}$: *Convexity*

Convex Shape



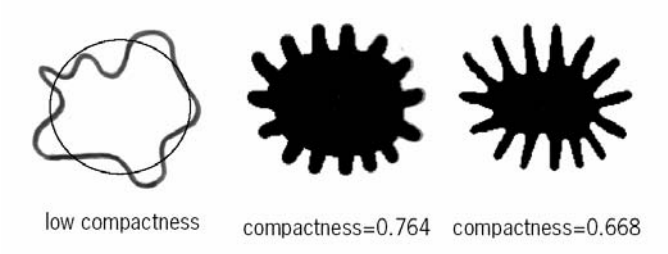
Convexity & Solidity



Compactness

- Feature relates to a standard shape: circle
- Descriptor for irregular boundaries
- Shape Factor a.k.a Form Factor
- $P_{circle} = 2\pi\ r$; and thus $(P_{circle})^2 = 4\pi^2\ r^2$
- $A_{circle} = \pi\ r^2$
- Shape Factor = $(P_{object})^2 / 4\pi\ A_{object}$;
- Circle: Shape Factor = 1
- The more irregular, the lower the value

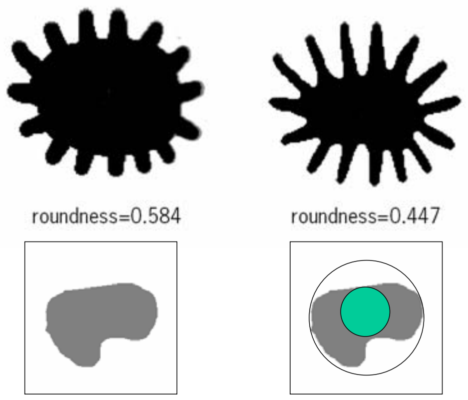
Shape Factor - Compactness



Roundness

- Roundness or Circularity is a also ratio of area to perimeter
- The irregularity are cancelled out by using the perimeter of the convex hull
- Roundness = $4\pi A_{object} / (P_{hull})^2$
- The degree to which the object approaches a sphere-shape is expressed by the sphericity:
  sphericity = $R_{inscibed} / R_{circumscribed}$

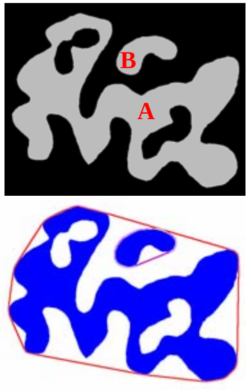
Roundness & Sphericity



Shape Measures

- Solidity
  - is a ratio of Surface Area
  - normalized
- Convexity
  - is a ratio of Perimeter Length
  - normalized
- Roundness
  - is a ratio of Surface Area over square Perimeter Length
  - compared to a circle
- Compactness
  - is a ratio of square Perimeter Length over Surface Area
  - compared to a circle

Example with known Shape



Area A	= 104607
Area A_{hull}	= 190882
Perimeter A	= 2691.5
Perimeter A_{hull}	= 1669.9
Solidity A	= 0.55
Convexity A	= 0.62
Roundness A	= 0.47
Area B	= 8438
Area B_{hull}	= 10138
Perimeter B	= 418,6
Perimeter B_{hull}	= 389,6
Solidity B	= 0.83
Convexity B	= 0.93
Roundness B	= 0.69

Curvature

- Curvature is a boundary descriptor
- Curvature (κ) represents the rate of change of direction at a particular point on that curve.
- K-slope: at boundary point (x_i,y_i) the k-slope is estimated from the slope op the line joining ($x_{i-k/2}$,$y_{i-k/2}$) and ($x_{i+k/2}$,$y_{i+k/2}$). Where 0<k<N, a positive small even integer value.

$$k_{slope} = \tan^{-1}\left(\frac{y_{i+k/2}-y_{i-k/2}}{x_{i+k/2}-x_{i-k/2}}\right)$$ Clockwise, horizontal slope =0

Curvature and Shape

- Curvature provides local shape information
  - Convex parts result in positive curvatures
  - Concave parts result in negative curvatures
- Curvature calculated from K-slope:

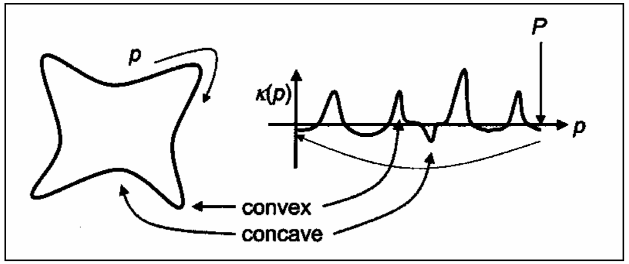
$$\kappa(\theta)=\left\{\tan^{-1}\left(\frac{y_{i+k}-y_i}{x_{i+k}-x_i}\right)-\tan^{-1}\left(\frac{y_i-y_{i-k}}{x_i-x_{i-k}}\right)\right\}\bmod 2\pi$$

- The total absolute curvature is given by:

$$\kappa_{TOTAL}(\theta)=\frac{1}{P}\sum_{p=1}^{P}\left|\kappa(\theta)\right| \qquad 2\pi\le\kappa_{TOTAL}\le\infty$$ P = Length

Plot of Local Curvature

- Local curvature is periodic over boundary length P



Bending Energy

- The Bending Energy E_c represents the Energy required to bend a rod to the desired shape.
- The Bending Energy is calculated from the curvature values over a boundary length P as: $E_c=\frac{1}{P}\sum_{p=1}^{P}(\kappa(\theta))^2 \qquad \frac{2\pi}{R}\le E_c\le\infty$
- Minimum value 2π/R is obtained for a circle of radius R

Fractals

- Fractal behavior is observed when a measurement is n-1 lower then viewing dimension n
- Examples:
  - Perimeter in n=2
  - Surface in n=3

Scale difference: 80x

Fractal Dimension

- A rough boundary will produce a different measurement at different resolutions. The higher the resolution the higher the measurement.
- Typical for a measurement that is one dimension lower than the measurement grid.
  - Perimeter in 2D
  - Surface area in 3D
- Fractal dimension is the rate of change (increase) of the object perimeter as the measurement is realized at a smaller scale (resolution)

Moments

- Moments are defined as:

$$m_{pq} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^p y^q f(x,y)\,dxdy$$

- For the discrete case this is formulated as:

$$M_{pq} = \sum_{0}^{sy-1}\sum_{0}^{sx-1} x^p y^q F(x,y)$$

- Centralized Moments are translation invariant

$$\mu_{pq} = \sum_{o}^{sx}\sum_{o}^{sy} (x-\bar{x})^p (y-\bar{y})^q \cdot F(x,y)$$

- Where $\bar{x} = \frac{M_{10}}{M_{00}}$ and $\bar{y} = \frac{M_{01}}{M_{00}}$

Moments and Shape Analysis

- F(x,y) can be binary function, geometry corresponds to shape of distribution function F(x,y)

$$M_{pq} = \sum_{0}^{sy-1}\sum_{0}^{sx-1} x^p y^q F(x,y)$$

- Used in shape description
- Moments use values up to x^4 and higher
  - Noise is amplified
  - Caution with interpretation

Centralized, Normalized Moments

- **Centralized moments**:
  - first order moments set to zero

$$\mu_{pq} = \sum_{0}^{sy-1}\sum_{0}^{sx-1}\left(x-\bar{x}\right)^p\left(y-\bar{y}\right)^q F(x,y)$$

- **Normalized moments** are derived from μ_{pq}

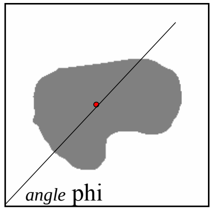
$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{\gamma}}, \text{ with } \gamma = \frac{p+q}{2}+1$$

  - rotation invariant
- From normalized moments [0,3] a number of 7 Moment Invariants are derived.

Eccentricity from Moments

- Using the centralized moments the eccentricity is computed form its second order moments:

$$ecc = \frac{\left(\mu_{20}-\mu_{02}\right)^2 + 4\cdot\left(\mu_{11}\right)^2}{\mu_{00}}$$



$$\bar{x} = \frac{M_{10}}{M_{00}} \qquad \bar{y} = \frac{M_{01}}{M_{00}}$$

Major-Minor Axis from Moments

- The semi-major axis is given by:

$$\alpha = \left(\frac{2\cdot\left[\mu_{20}+\mu_{02}+\sqrt{\left(\mu_{20}-\mu_{02}\right)^2+4\cdot\left(\mu_{11}\right)^2}\right]}{\mu_{00}}\right)^{\frac{1}{2}}$$

- The semi-minor axis is given by:

$$\beta = \left(\frac{2\cdot\left[\mu_{20}+\mu_{02}-\sqrt{\left(\mu_{20}-\mu_{02}\right)^2+4\cdot\left(\mu_{11}\right)^2}\right]}{\mu_{00}}\right)^{\frac{1}{2}}$$

Orientation from Moments

- The orientation of the major principal axis with respect to the x-axis of the image is given by:

$$\varphi = \tfrac{1}{2}\tan^{-1}\left(\frac{2\mu_{11}}{\mu_{20}-\mu_{02}}\right)$$

- derived from second order central moments

Moment Invariants

- From the normalized moments (0th to 3rd order) a set of 7 invariants can be calculated; the first two results are :

$$\phi_1 = \eta_{02} + \eta_{20} \qquad \phi_2 = (\eta_{02} - \eta_{20})^2 + 4\eta_{11}^2$$

- From φ_1 and φ_2 the following parameters are calculated:

$$\lambda_1 = 2\pi(\phi_1 + \sqrt{\phi_2})$$

$$\lambda_2 = 2\pi(\phi_1 - \sqrt{\phi_2})$$

Invariant-based shape descriptors

Extension (E),
Dispersion (D) and
Elongation (L), given by:

$$E = \log_2 \lambda_1 = c \ln \lambda_1 \cong \ln \lambda_1$$

$$D = \log_2 \sqrt{\lambda_1 \lambda_2} = \frac{1}{2} c \ln \lambda_1 \lambda_2 \cong \frac{1}{2} \ln \lambda_1 \lambda_2$$

$$L = \log_2 \sqrt{\frac{\lambda_1}{\lambda_2}} = \frac{1}{2} c \ln \frac{\lambda_1}{\lambda_2} \cong \frac{1}{2} \ln \frac{\lambda_1}{\lambda_2}$$

Extension

Extension: $E = \log_2 \lambda_1 \Leftrightarrow$

$E = c \ln \lambda_1 \cong \ln \lambda_1$

• measures how much the shape differs from the circle.

• is *zero* for circular shape

• increases without upper limit as the shape becomes less compact

Dispersion

Dispersion: $D = \log_2 \sqrt{\lambda_1 \lambda_2} \Leftrightarrow$

$D = \frac{1}{2} c \ln \lambda_1 \lambda_2 \cong \frac{1}{2} \ln \lambda_1 \lambda_2$

- minimum extension that can be attained by uniform compression of the shape.
- to minimize its extension the shape must be compressed along long axis of the shape.
- i.e. long axis is the unique axis
- invariant to stretching, compressing or shearing the shape in any direction

Elongation

Elongation: $L = \log_2 \sqrt{\frac{\lambda_1}{\lambda_2}} \Leftrightarrow$

$$L = \frac{1}{2} c \ln \frac{\lambda_1}{\lambda_2} \cong \frac{1}{2} \ln \frac{\lambda_1}{\lambda_2}$$

- measures how much the shape must be compressed along its long axis in order to minimize the extension
- never less than zero
- never greater than extension

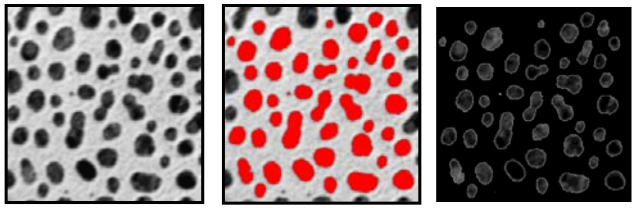
No	Extension	Dispersion	Elongation
1	0,1197	0,0542	0,0655
2	0,3998	0,0524	0,3474
3	0,7575	0,0550	0,7025
4	0,8725	0,1740	0,6985
5	0,0920	0,0262	0,0659
6	0,3784	0,0277	0,3506
7	0,7411	0,0313	0,7099
8	0,8591	0,1434	0,7157
9	0,0816	0,0138	0,0679
10	0,3617	0,0160	0,3457
11	0,7243	0,0199	0,7044
12	0,8350	0,1029	0,7321

Purdue University, Dept. Cytochemistry

Density

- The average density of a binary object is 1
- A binary object is a mask for the density image.
- The average density within the object is the ratio of two zero order moments
- Using the object as a mask one can also obtain the integrated density, which is the zero order moment of the mask

Using object intensity

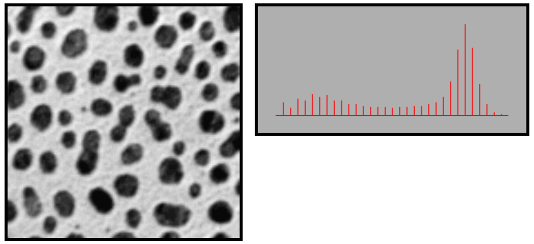


- Intensity, Density, Integrated Density, Object Density.
- Measurement are done taking background into account
- Meaning different per type of microscopy
- Using Lambert-Beer law

Doing measurements

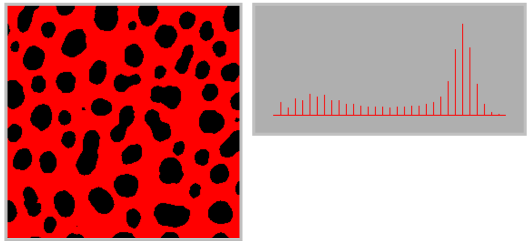
- Precision:
  - same result after repeated application: reproducibility
- Accuracy:
  - accurate result, close to the true value (no bias)
  - measure from standard calibrated caliper
- Tools:
  - standard deviation, measure in object units $\sigma = \sqrt{\frac{\sum [x_i - \mu]^2}{N}}$
  - Coefficient of variation, relative standard deviation $cv = \frac{\sigma}{\mu} * 100\%$

Image Analysis trajectory

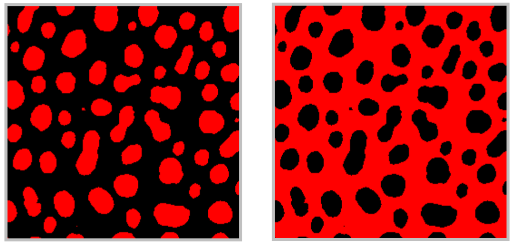


A simple test-image: cermet

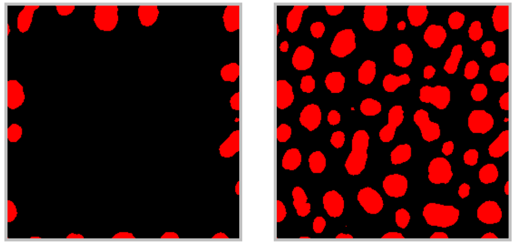
Brightness Threshold (isodata)



Invert: object to foreground

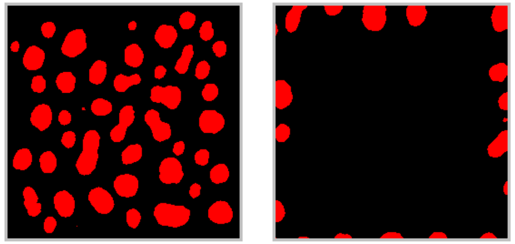


Find objects on image border



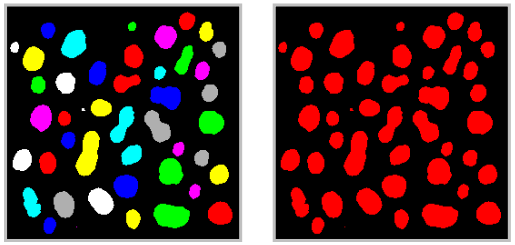
Propagation, from border with 4-connected components

Remove objects on image border



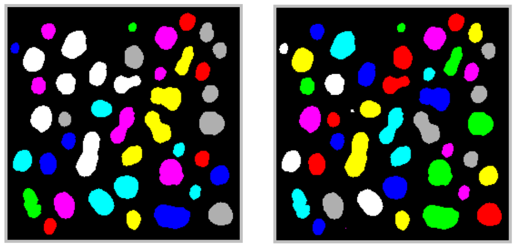
Xor result with original

Label the image



Label the objects in the image

Select relevant objects



Remove irrelevant objects

Measure objects: Results

Object	Size	Mean	GravX	GravY	CenterX	CenterY	MaxX	MaxY	MinX	MinY	MaxVal	MinVal	Fmax	Fmin	FmaxP	InertM1	InertM2	Peri	Bend
	A.U.^2	ADU	A.U.	A.U.	A.U.	A.U.	A.U.	A.U.	A.U.	A.U.	ADU	ADU	A.U.	A.U.	A.U.	A.U.^2	A.U.^2	A.U.	A.U.^2
74	74	91.351	137.072	21.737	137.041	21.73	141	26	133	17	120	56	11.294	9	10	6.925	4.987	432.265	1.005
92	92	87.913	7.935	44.655	7.826	44.75	12	50	3	39	120	56	12.705	9.869	12.698	9.516	5.685	81.429	1.032
149	149	68.94	167.331	72.846	167.43	72.732	174	80	162	66	120	32	15.765	13	15	14.549	9.785	104.529	1.057
158	316	69.848	197.247	179.678	197.089	179.604	212	211	182	149	120	24	17.155	12.644	17.006	635.412	12.389	108.691	1.091
184	184	71.826	62.671	122.796	62.505	122.81	69	131	56	115	120	24	17.123	14	17	17.061	12.604	116.772	1.038
201	201	68.06	46.29	240.335	46.179	240.408	53	249	40	232	120	32	19.027	14	18	21.79	11.773	121.364	1.112
216	216	56.333	233.331	47.767	233.176	47.023	240	56	225	39	120	8	18.72	16	18	20.27	14.778	127.873	1.099
221	221	64.796	44.599	26.382	44.498	25.986	52	35	37	18	120	16	18.49	16	18	20.093	15.537	127.199	1.074
228	228	61.333	67.007	146.882	66.873	146.592	74	156	59	138	120	24	19.438	15.575	19.355	22.236	14.95	131.404	1.091
233	233	66.197	213.884	166.97	213.751	166.339	221	175	206	158	120	8	19.027	16	18	21.431	16.225	130.587	1.11
245	245	73.469	34.027	86.087	33.641	85.996	41	95	26	77	120	16	19.786	16	19	23.317	16.419	134.974	1.095
257	257	52.109	214.358	70.419	214.183	70.502	222	80	206	60	120	8	21.879	15.796	21.857	27.952	15.031	140.607	1.135
264	264	78.667	218.974	27.645	218.742	27.678	226	38	211	17	120	32	22.586	16	22	30.444	14.784	144.015	1.224
267	267	62.981	138.33	233.376	138.06	233.206	146	244	131	223	120	24	22.213	16	22	28.8	15.802	141.75	1.153
272	272	64.382	222.918	95.159	222.599	94.849	231	104	214	86	120	16	20.722	17.595	20.685	25.113	18.711	140.546	1.075
273	273	48.352	197.504	15.901	197.355	15.85	206	25	189	6	120	8	20.645	17.786	20.615	24.779	19.131	141.301	1.066
350	350	56.617	102.635	111.457	102.837	111.326	114	120	92	102	120	24	23.802	19	23	34.336	22.683	160.196	1.125
364	728	51.505	126.162	177.29	138.562	178.144	243	195	35	160	120	8	25.038	19	24	8.98E+03	34.011	165.012	1.163
387	774	73.974	166.262	110.08	165.616	111.062	204	174	125	43	120	24	34.613	14.748	34.613	3.60E+03	19.101	199.744	1.917
398	398	53.307	16.224	168.143	16.123	168.726	26	180	6	157	120	8	26.238	20.639	25.631	39.78	25.386	173.337	1.169
399	399	67.749	64.337	84.226	63.925	83.807	74	95	53	73	120	32	24.77	22	23	35.299	28.821	173.786	1.147
407	814	53.877	114.406	77.781	114.466	78.342	146	94	90	60	120	8	27.627	19	27	300.489	40.987	179.827	1.275
432	432	43.963	138.321	55.301	138.567	54.725	149	67	129	42	120	8	26.493	21	26	42.209	28.162	180.969	1.176
450	450	86.916	26.577	216.834	26.091	215.42	36	231	17	199	120	24	34.241	17.557	33.894	77.056	17.412	213.435	2.219

Contents (2)

- Ratio Imaging
- 3D measurements
  - Connectivity
  - Kernels
- 3D features
  - Volume
  - Surface Area

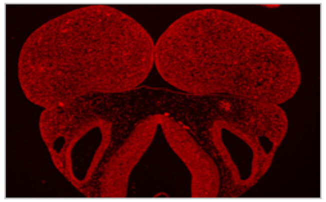
Adding more dimensions

- In general 3D imaging is referred to as imaging with 3 spatial dimensions.
- Sampling is in xy-plane and along z-axis.
- Multi-dimensional imaging can also refer to the time axis which is added: (x,y,t)
  - Time dependent phenomena; Time resolved
  - Time lapse, Time-lapse microscopy
  - e.g. Biochemical reactions, interactions: FRET
  - e.g. Physiological reactions, Ca++ imaging

Ratio Imaging

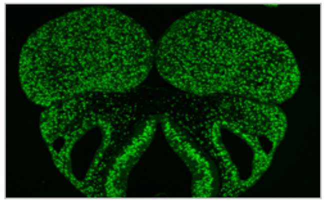
- Densities of two states
  - over time
  - in space
- Result shows areas where difference is found.
- From the ratios a spatial map can be derived
- Typically applied with Fluorescence Microscopy
- Output often used for visual inspection
- Techniques: FRET,FRAP, FLIM (x,y,t)

Ratio Imaging: Phenotype Analysis



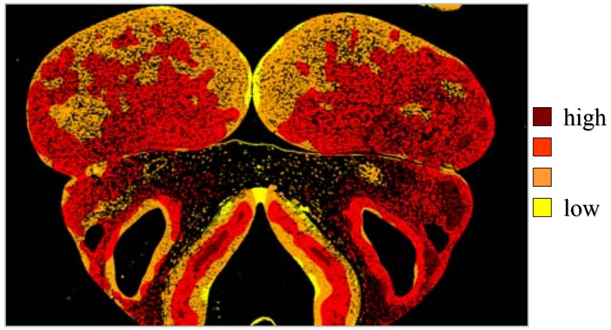
Bopro

Mouse embryo jaw



BrdU

Proliferation Topography

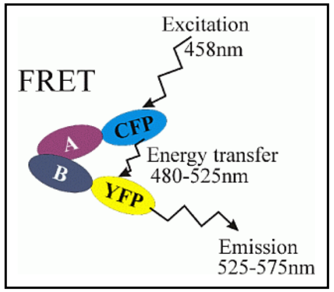


Molecular Interactions in Time

- FRET: fluorescence resonance energy transfer
- Complex formation of proteins
  - Energy transfer occurs only if proteins are in close proximity
  - Complex formation: 10-100 Å, which is 1-10 nm
- Create conditions for complex formation
  - experimental setup
  - physiological conditions of cells
  - radiation less energy transfer
  - proteins tagged with fluorescent proteins (GFP)

Illustration of FRET principle

- 50% efficient energy transfer: Förster Radius R0
- FRET aka FET
  Förster Energy Transfer
- R0 dependent on Chemical properties of reactant and Fluorescent molecule

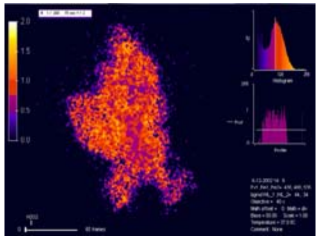
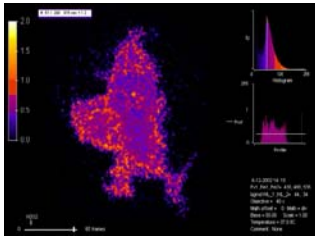


Example FRET Imaging

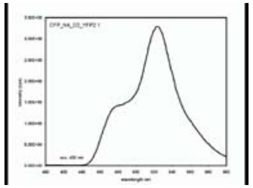
- Complex exists at t=0
- Complex bound to membranes
- Redox reaction: dissociation of the complex

- Normal: energy transfer form
  - CFP(480 nm) to YFP (525nm)
  - CFP-sensor-YFP
- Administer reagent (peroxide) at t=n
- Complex dissociates
  - energy transfer diminishes quickly

Screenshots of the movies

start



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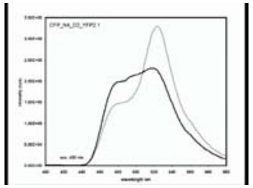


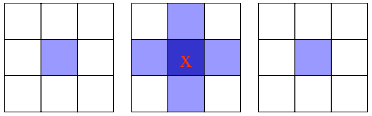
Spatial 3D imaging

- 3 dimensions: (x,y,z)
  - Sampling in xy
  - Sampling in z
- 3 dimensions
  - Sampling in yz,φ
  - Reconstruct to x,y,z
  - Tomography, Optical Projection Tomography
  - Radon transforms
- 3D, plan parallel
  - CLSM, MphM
  - Serial sections

Connectivity in 3D

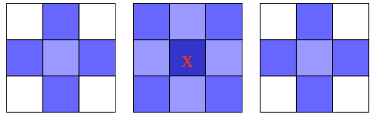
- In 2D the connectivity amounts to
  - 4 connected
  - 8 connected
- In 3D the connectivity expands to
  - 6 connected, only direct neighbors
    - One facet in common (face connectivity)
  - 18 connected, diagonal neighbors
    - One edge in common (edge connectivity)
  - 26 connected, far diagonal neighbors
    - One corner-point in common (vertex connectivity)

6 Connected



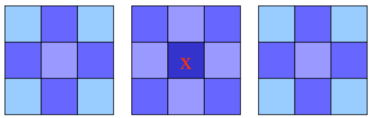
Plane z-1 z z+1

18 Connected



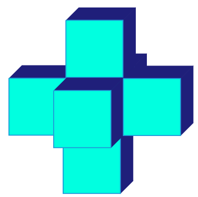
Plane z-1 z z+1

26 Connected

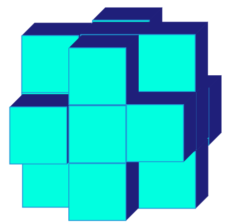


Plane z-1 z z+1

3D connectivity

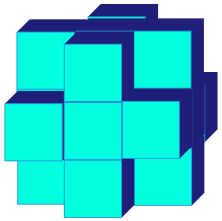


6-connected

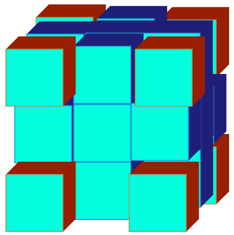


18-connected

3D connectivity



18-connected



26-connected

Isotropy

- Sampling in pixels assumes isotropy in the plane: square pixels
- Sampling in voxels assumes isotropy in the space: cubical voxels
- Connectivity remains identical under

non-isotropic voxels, inaccuracy increases.

Filters & Kernels

- 2D transposes to 3D case
- More connectivity
  - Mathematical morphology
  - Labeling
  - Convolutions, voxels taken into account?
- More complexity
  - More computations
  - Isotropy - Anisotropy

3D Measurements: Volume

- Volume (voxel counting)

$$m_{pqr} = \iiint x^p y^q z^r f(x,y,z)\, dxdydz$$

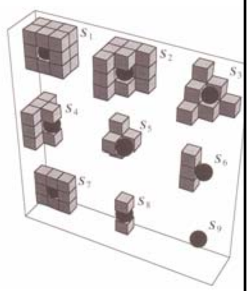
$$\mu_{pqr} = \sum_o^{sx} \sum_o^{sy} \sum_o^{sz} (x-\bar{x})^p (y-\bar{y})^q (z-\bar{z})^r \cdot F(x,y,z)$$

- Axis length, Shape extension
  - From 2nd order moments
- Shape
  - Using convex hull volume
  - Convex deficiency

Surface Area in 3D

Surface Area

- Choose a connectivity
- How to describe a plane?



- Determine the number of facets that are on the boundary: voxel classification
- Weight factors for voxels W_1, W_2, W_3
- Smooth surface: sampling (isometric)

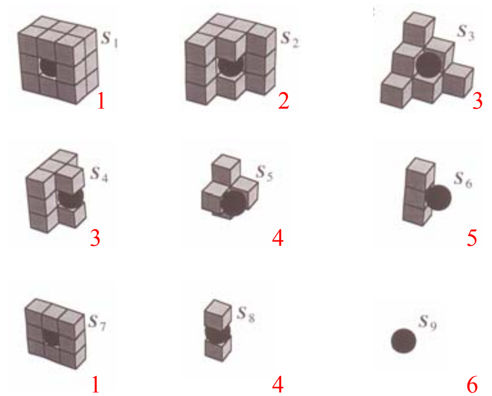
Figure. From thesis J.C. Mullikin

6-connected evaluation

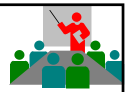
regular

curved

plane,line, point



Review Lecture 13



- Measurement
- Surface Area
- Perimeter
- Shape, range of features
- Curvature
- Density and Ratios
- 3D measures

Acknowledgements

Books

- Gonzales & Woods: Digital Image Processing
- FJV

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- I.T.Young, J.M. Mullikin

Pictures

- FJV, Experimental data
- Univ. Guelph, Purdue Univ.